

Control Theory Exercises

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Control Theory Exercises

1 State-Space Systems

1. Consider the system

$$\dot{x}(t) = ax(t) + bu(t).$$

Find the solution for arbitrary initial condition $x(0) = x_0$.

2. Determine the stability of the system

$$\dot{x} = -3x.$$

3. Determine the equilibrium points of

$$\dot{x} = x - x^3.$$

4. Solve the system

$$\dot{x} = Ax, \quad A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}.$$

5. Compute the matrix exponential e^{At} for

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}.$$

2 Controllability and Observability

6. Determine whether the following system is controllable:

$$\dot{x} = Ax + Bu,$$

with

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

7. Determine whether the following system is observable:

$$\dot{x} = Ax, \quad y = Cx,$$

where

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

8. Compute the controllability matrix of

$$A = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

9. Compute the observability matrix of

$$A = \begin{pmatrix} 0 & 1 \\ -4 & -5 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

10. Determine whether the pair (A, B) is stabilizable.

11. Determine whether the pair (A, C) is detectable.

3 Transfer Functions and Laplace Transforms

12. Find the transfer function of the system

$$\ddot{y} + 3\dot{y} + 2y = u.$$

13. Find the poles and zeros of

$$G(s) = \frac{s+1}{s^2+4s+3}.$$

14. Determine whether the system

$$G(s) = \frac{1}{s^2+s+1}$$

is stable.

15. Find the Laplace transform of

$$f(t) = e^{-2t} \sin t.$$

16. Compute the inverse Laplace transform of

$$F(s) = \frac{1}{s(s+1)}.$$

17. Find the impulse response of

$$G(s) = \frac{1}{s+2}.$$

18. Find the step response of

$$G(s) = \frac{5}{s+5}.$$

4 Classical Control

19. Sketch the root locus of

$$G(s)H(s) = \frac{K}{s(s+2)}.$$

20. Use the Routh–Hurwitz criterion to determine stability of

$$s^3 + 2s^2 + 3s + 4 = 0.$$

21. Determine the gain K such that the closed-loop system is stable:

$$G(s) = \frac{K}{s(s+1)(s+2)}.$$

22. Determine the frequency response of

$$G(s) = \frac{1}{s+1}.$$

23. Sketch the Bode plot of

$$G(s) = \frac{10}{s(s+2)}.$$

24. Determine the Nyquist plot of

$$G(s) = \frac{1}{s+1}.$$

25. Determine the gain margin and phase margin of

$$G(s) = \frac{1}{s(s+1)}.$$

26. Design a proportional controller for

$$G(s) = \frac{1}{s+1}.$$

27. Design a PI controller for a first-order system.

28. Design a PID controller for

$$G(s) = \frac{1}{s(s+1)}.$$

29. Determine the steady-state error for a unit step input.

30. Determine the steady-state error for a ramp input.

5 Linear Algebra and Canonical Forms

31. Determine the eigenvalues of

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

32. Determine whether the system is asymptotically stable:

$$\dot{x} = Ax, \quad A = \begin{pmatrix} -1 & 2 \\ -3 & -4 \end{pmatrix}.$$

33. Find a state feedback law

$$u = -Kx$$

that places the poles at -2 and -3 .

34. Find the controllable canonical form of

$$G(s) = \frac{s+2}{s^2+3s+2}.$$

35. Find the observable canonical form of

$$G(s) = \frac{1}{s^2+2s+1}.$$

36. Compute the Jordan canonical form of

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

6 Nonlinear Systems and Lyapunov Stability

37. Consider the pendulum equation

$$\ddot{\theta} + \sin \theta = u.$$

Linearize the system around $\theta = 0$.

38. Linearize the nonlinear system

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -x_1^3 + u.$$

39. Determine the phase portrait of

$$\dot{x} = x(1-x).$$

40. Solve the Lyapunov equation

$$A^T P + P A = -Q.$$

41. Determine whether

$$V(x) = x^T P x$$

is a Lyapunov function.

42. Analyze the stability of

$$\dot{x} = -x + x^3.$$

43. Determine the invariant sets of

$$\dot{x} = -x.$$

44. Apply LaSalle's invariance principle to a nonlinear system.

7 Optimal Control

45. Solve the Riccati equation associated with the scalar system

$$\dot{x} = ax + bu.$$

46. Compute the controllability Gramian of a linear system.

47. Compute the observability Gramian of a linear system.

48. Solve the optimal control problem

$$J(u) = \int_0^T (x^2 + u^2) dt.$$

49. Derive the Hamiltonian equations in Pontryagin's maximum principle.

50. Solve the linear quadratic regulator problem for

$$\dot{x} = Ax + Bu.$$

8 Discrete-Time Systems

51. Consider the discrete-time system

$$x_{k+1} = Ax_k + Bu_k.$$

Determine conditions for stability.